

PH 504: Organic Optoelectronics

Course Code: PH 504

Course Name: Organic Optoelectronics

L-T-P-C: 3-0-0-3

Approval: 2nd Senate

Course Contents

PART I

- **Organic Molecules**
 - Electronic structure of atoms, Atomic and Molecular Orbitals, LCAO, Bonding and antibonding orbitals, Covalent Bond, Sigma and Pi Bonds, Energy Levels, Spectroscopic properties. [4 Lectures]
- **Photophysics of Molecules and Aggregates**
 - Excited states: Absorption and emission, Singlet and triplet states, Radiative and non-radiative transitions, Aggregates, Vander Waals Bonding, Hydrogen Bonding, Dimer, Eximers. [2 Lectures]
- **Excitons**
 - Wannier Exciton, Charge-transfer Exciton Frenkel Exciton, Exciton Diffusion, Excitonic Energy Transfer. [2 Lectures]
- **Conduction in Organic Solids**
 - Conductivity: carrier concentration versus mobility, Carrier generation, Hopping transport, Mobility measurements, Traps. [2 Lectures]
- **Photovoltaics and Photodetectors**
 - Photovoltaic Devices: Organic Heterojunction Photovoltaic Cells, Organic/Nanorod hybrid Photovoltaics, Gratzel Cells (Dye sensitized solar cells), Photodetector Devices. [5 Lectures]
- **Organic Light Emitting Devices**
 - Basic OLED Properties, Charged Carrier Transport, Organic LEDs, Quantum Dot LEDs. [8 Lectures]
- **Lasing Action in Organic Semiconductors**
 - Lasing Process, Optically Pumped Organic Lasers, Electrical Pumping of Organic Lasers. [2 Lectures]
- **Organic Thin Film Transistors**
 - OFETs: Materials, Contacts, Applications, Nanotube Transistors. [2 Lectures]
- **Device Fabrication Technology**
 - Growth Techniques: Evaporation, Langmuir-Biodget, Chemical Vapor Phase Deposition, Ink-Jet Printing, Self Assembly. [3 Lectures]

PART II

Project

- Literature review on a certain relevant topic. [10 Lectures]

Text Books:

No textbook required. Lecture notes and handouts will be provided.

General References:

1. Gilbert & Baggott, **Essentials of Molecular Photochemistry**, CRC Press, 1991.
2. K. K. Rohatgi, Mukherjee, **Fundamentals of Photochemistry**, New Age International, 1978.
3. Pope & Swenberg, **Electronic Processes of Organic Crystals and Polymers**, 2nd Edition, Oxford University press, 1999.
4. H. Meier, **Organic Semiconductors**, Verlag Chemie GmbH, 1974.
5. Wolfgang Brütting, **Physics of Organic Semiconductors**, John Wiley, 2005.